Technical review

Item #5 · leaf-expired.crt · Run ID: 20261113T232802Z-2f0fe5

Path	`R:\evidence\certs\leaf-expired.crt`
Size (bytes)	1574
Last modified (UTC)	2026-11-12T06:00:00Z
Extension	.crt
Detected type (signature)	X.509 certificate
Status	Problem found
Case	RV-SAMPLE-001
Evidence ID	E0005
Case registry	Verified: identical to the registered evidence
Registered at intake (UTC)	2026-11-13T23:27:52.795Z

Hashes

sha256	`346e22485a3cb93677fa1262bd12c5870f9d656d4737004018c34c8bcf18e0c9`
md5	`efebfe58df5781bd87e2eb96df0b1595`

Findings

[High] Certificate expired at the date checked (CERT_EXPIRED_AT_DATE)

What it shows	C=CO, O=Rakvor Lab, CN=expired.lab.rakvor.test was valid until 2025-01-01T00:00:00Z; the chain was checked at 2026-06-01T00:00:00Z, after that date.
What it does not prove	That a signature made with it is invalid: what matters for a signature is the validity at signing time (check again with --at signing-time or the date of signing). Expiry of old certificates is normal.

Method

Each file was opened read-only, hashed in a single pass with the selected algorithms, and checked against a table of file signatures. Every action was written to a hash-chained audit log.

Audit trail

Audit entries	#22 – #25
Log folder	`R:\case\runs\rv-certcheck\20261113T232802Z-2f0fe5\logs`
Digital signature	This PDF is not digitally signed.

Limitations

- Every chain is judged at one instant (--at, recorded in each report); the same certificate can be valid at one date and expired or revoked at another.
- Offline, revocation is checked only against the CRLs and OCSP responses supplied (--crl, --ocsp, or found among the evidence); a revocation published elsewhere is not seen.
- Trust lists are used as dated snapshots (date and SHA-256 recorded); services added, suspended or withdrawn after the snapshot are not reflected.
- In version 0.1 the trust list's own XAdES signature is not verified; the snapshot's source and SHA-256 are recorded so it can be checked independently. The Colombian list is a curated file, because ONAC publishes no

machine-readable list.

- Certificates issued by Rakvor's internal PKI are for internal use only: they are not a firma digital under Ley 527 de 1999, art. 28 (Colombia), and not qualified certificates under eIDAS (Regulation (EU) No 910/2014).
- The tool's results have been compared with OpenSSL and Python reference implementations on test sets, but no error rate has yet been measured on a validated corpus.
- A result without findings does not prove that a file is authentic or unaltered before acquisition.
- Modification times come from the file system of the examined copy and may differ from the original source.
- A trusted timestamp of the audit log shows that the log existed in this exact form at the stated time; it does not show that the examined copy matches the original source.